



Research Article

Instability monitoring of molten pool in pure copper laser welding based on a multi-scale cascade model and spatial optical signals

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ABSTRACT

During laser welding of pure copper, instabilities such as violent molten pool oscillation, large spatters, and melt ejection severely damage weld quality, which are caused by copper's high reflectivity on commonly used infrared laser. The seriously unstable molten pool and keyhole and complicated laser-material interactions result in complex process signal emission waveforms, adding to the difficulties in process stability monitoring tasks. In this work, to break down different contents in the complex signals and deeply analyzing signal-process relation, combinative spatial optical sensor system was designed, and time-frequency signal analysis in multi-scale windows was performed. It was found that the infrared radiation at the front and end side of molten pool indicates the oscillation behavior of liquid metal surface, and the signal fluctuation patterns of visible radiation from different height of metal vapor varied when meeting severe instability like melt ejections. Signal features were extracted based on the understanding of process mechanism and signal behaviors. A cascade model combining Artificial Neural Network (ANN) and Support Vector Machine (SVM) was introduced to predict weld seam quality, where the ANN model focused on short-time stability status perception and the SVM model was used to decide macroscopic seam formation defects based on combining outputs of the ANN model in a long-term sampling window. Application results showed that the recognition accuracy of pit was 100 % and the accuracy of uneven toe reached 86.3 %. The multi-source signals of unstable molten pool recognized by the cascade model were summarized. The evolution process of copper molten pool ejection was revealed.

1. Introduction

Pure copper has been widely used in the electric vehicle production due to its distinguished electrical conductivity [1]. The laser welding of pure copper has been increasingly concerned due to the small distortion, excellent flexibility, and high efficiency [2]. However, there is a significant difficulty in laser welding of pure copper. At room temperature, the absorption of pure copper for near infrared (NIR) laser is lower than 3 %, making it difficult in creating reasonable penetration depths in welding [3], resulting in a requirement of high laser energy input [4]. The increase of laser heat input causes serious instabilities in welding process [5]. Research shows that such instabilities are relative to the absorption rate increasing to about 10 % when near melting point [6]. The significant increase of absorption when temperature rises results in instability issues of keyhole, causing spatters [7], melt ejections [8] and porosity defects [5]. Among them, severe spatters and melt ejections seriously deteriorate seam quality and urgently needs to be resolved.

During research on this topic, observation by in-situ X-ray imaging

showed that large bulges existed at the times of melt ejection formation [9]. Zhang et al. [10] analyzed different energy input settings in the welding of T2 copper and drew a conclusion that the bulging effect was due to the abnormal energy concentration at keyhole bottom in the welding process of high-reflective materials. Therefore, two routes of solving the process instability problem can be introduced. On the one hand, owing to the higher absorption rate of laser energy in visible light range for pure copper, green or blue laser sources can be used to decrease the power needed in welding process [11]. Hess et al. [12] found stability increasing in copper laser welding by using green laser source of 515 nm wavelength. Hummel et al. [13] studied the influence of various wavelengths of laser on the welding stability of pure copper. It was proved that the stability of welding process should also deserve attention at the wavelength with higher absorption. Maina et al. [14] designed a multi-laser welding strategy containing both green and infrared laser to obtain high-penetration and stable welding process. However, even with green laser sources there were still reports of instabilities, indicating that the instabilities could be caused by other

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factors such as plume [15] or laser beam quality [16]. On the other hand, techniques controlling energy input distribution or alleviating pressure accumulation in keyhole bottom are also possible ways to release the problem. Reisgen et al. [17] tried to decrease keyhole pressure by performing welding process in vacuum environment and discovered the welding process was profoundly stabilized. Ning et al. [18] analyzed mechanism of process instability for laser welding of high-reflective materials and performed power modulation to stabilize process. Franco et al. [5] improved process stability by using different laser oscillation mode and parameters to change energy distribution geometry. Zhang et al. [19] proved that the adjustable ring mode laser was helpful to improve the weld quality of copper laser-welded joints. Kaufmann et al. [20] compared laser beams with different intensity distribution shapes. Chianese et al. [21] confirmed the influence of laser shapes using numerical simulation technology. However, the above methods require complicated equipment and introduce multiple processing parameters that need to be tuned. At the same time, influence factors like oxygen content in the materials [22], composition of metal plume [23], or even laser spot diameter [16] make the controlling target even more complex. The sensitivity of interaction characteristics between pure copper and laser makes it difficult to strictly control the stability of welding process. Methods like experiments and numerical simulation are suitable to prediction such process in scientific research backgrounds [24] but lack the efficiency and adaptivity needed in industry application. From this perspective, process monitoring is still crucial both to understand the instability mechanism and to detect defective joints.

Recently, process monitoring technology based on data mining and machine learning models developed rapidly in the field of material science and manufacturing [25]. In welding, process signals are collected and used as data to build machine learning models [26]. Signal types in use contain acoustic signals [27], electrical signals [28] and optics signals [29]. In these types of signals, optical radiation signals are the most utilized in laser welding, due to the easy-to-build acquisition equipment, fast sampling rate and highly informative signal waveforms [2]. Sensors with different photosensitive wavelengths also contain different process information [30], which is suitable to build a data-driven system for complicated monitoring target like pure copper laser welding. However, although profound progress has been made in the field of stainless steel or aluminum alloys [31], there are still few reports on the process monitoring and defect prediction methods for pure copper laser welding, which might be caused by extremely fluctuated signals and exceedingly unstable laser-copper interaction. The relation between signal shapes and process states, as well as the method to extract useful signal information from the violent fluctuation waveforms requires to be established.

In this work, a process monitoring system with an intelligent defect prediction model for pure copper laser welding is proposed, based on in-depth analysis of process optics signals. Optics radiation signals at different spatial positions with different wavelength ranges were collected. The visible light radiation of metal vapor plume is collected at the high position and at the low position respectively. The infrared light radiation of molten pool is collected at the front-end and at the back-end respectively. Four photoelectric sensors were collected synchronously to observe the status of metal vapor and molten pool was detected. A cascade model combining an Artificial Neural Network (ANN) and a Support Vector Machine (SVM) was designed. The ANN was used in a short-time window for fast perception. The SVM was executed in a long-time window for status recognition. Application results showed that the cascade model has good capability to recognize the instable phenomena causing uneven weld toes and pits. The multi-source signals of instable process were summarized. The evolution process of copper molten pool ejection was revealed.

2. Experimental procedure

The T2 copper plate with 5 mm thickness was used in the laser welding. The alloying element composition of T2 copper was presented in Table 1. A Raycus RFL-4000/2000 adjustable ring mode (ARM) laser source was used in the laser welding, which was able to output continuous laser with a wavelength of 1080 ± 5 nm. The maximum power of core laser and ring laser is 4000 W and 2000 W, respectively. The focus diameter of core laser is 50 μm , and the diameter of ring is 150 μm . Four groups of welding experiments were carried out, as shown in Table 2. Different speeds and laser power values are designed to extend the fitness of the monitoring model on different situations. The welding speed of first three group A was 3 m/min, while the Group D was 6 m/min. Each set of parameters was weld 5 times repeatedly. The power ratio of the core laser and the ring laser is fixed at 2. The incident angle of laser beam was set to 5° towards welding direction. There was no defocus amount for both the core laser and the ring laser.

To enhance the coverage and separability of collected signals, spatial and multi-wavelength optics sensing system was built. During laser welding, the two major optics sensing targets are the molten pool and metal vapor. The model pool is liquid metal with temperature higher than liquidus temperature and lower than evaporation point of the base metal. For pure copper, the molten pool temperature mostly lies in the range of 1357–2843 K according to data from National Institute of Standards and Technology (NIST) and was confirmed in our experiment using infrared thermal imaging camera. Optical radiation in short-wave infrared (SWIR, 900–1700 nm) region is commonly used to detect temperature information of this range. The light emission of laser vapor comes from the ionized metal plasma formed in very high temperature caused by laser energy input [32]. Optics signals of metal plasma mostly consist of emission spectrum of metal elements and lies in the range of visible light range for copper, confirmed in spectrum data in experiment. Fig. 1(b) and (c) shows both experimental spectrum data collected from metal vapor and molten pool during welding process using spectrometer from Ocean Optics. Based on the above analysis, the laser welding process was monitored using two photodiodes in visible band (VI), two photodiodes in infrared band (IR), and two high-speed cameras, as shown in Fig. 1(a). All photodiodes are equipped with collimators and optical fibers for collecting light radiation in designated areas. The fiber and collimators limited sensing range of photodiodes in to circles with a diameter of 10 mm. Laser pointers were attached to fiber ends to access pointing direction of optical fibers. The sensing ranges and laser pointer directions were pre-calibrated before building the sensing system. And at system building, with the help of collimators, laser pointers, and fiber end fixtures, the sensing positions and ranges were adjusted and fixed. Fig. 1(d) shows the photo and schematic diagram of a sensor probe. The wavelength range of visible band is 330–900 nm, mainly from the optical radiation of vapor plume. A VI photodiode focused on the incident point of laser beam on the workpiece surface, aiming at collecting signals at keyhole opening. The focus of the other VI photodiode is located 5 mm above the workpiece, which ensured no overlap with the lower VI sensor. The position of VI sensors follows the following criterion: (1) The sensing range of the lower sensor should be centered at keyhole opening. (2) The height of the upper sensor is reduced as much as possible without overlapping the measurement area with the lower sensor. When there is a major change in laser equipment or welding parameter settings and a significant drop of model accuracy is found, the position of sensors needs to be recalibrated. According to the placement height, the VI photodiode at nether position is abbreviated as VI_N , and the VI photodiode at upper position is referred to as VI_U . As vapor plume sprayed out

Table 1
Alloying element composition of T2 copper (wt%).

Cu	Bi	Sb	As	Fe	Pb	S
>99.90	<0.001	<0.002	<0.002	<0.005	<0.005	<0.005

Table 2

Laser welding parameters.

No.	Core Power (W)	Ring Power (W)	Weld Speed (m/min)
A	2000	1000	3
B	3000	1500	3
C	4000	2000	3
D	4000	2000	6

of the keyhole in normal welding states, the VI_N and the VI_U detected the light intensity peak synchronously. The wavelength range of infrared band is 800–1750 nm, mainly containing the wavelength range of thermal radiation of molten pool. An IR photodiode focused on the

incident point of laser beam on the workpiece surface, pointing from the front-end of molten pool. The focus of the other IR photodiode was located 5 mm behind the incident point along the welding direction, pointing from the back-end. According to the placement position, the IR photodiode at the front end is abbreviated as IR_F , and the IR photodiode at the rear position is referred as IR_R . As the molten pool oscillated, the light intensity detected by the two IR photodiodes fluctuated accordingly. For each photodiode, the sampling speed of the light intensity signal was 50 kHz. In addition, two Phantom VEO710 high-speed cameras were used to observe the welding zone from different directions during the welding processes. One camera captured the images of the vapor plume from a horizontal direction, the other camera captured the images of the molten pool with a 45° depression angle. The

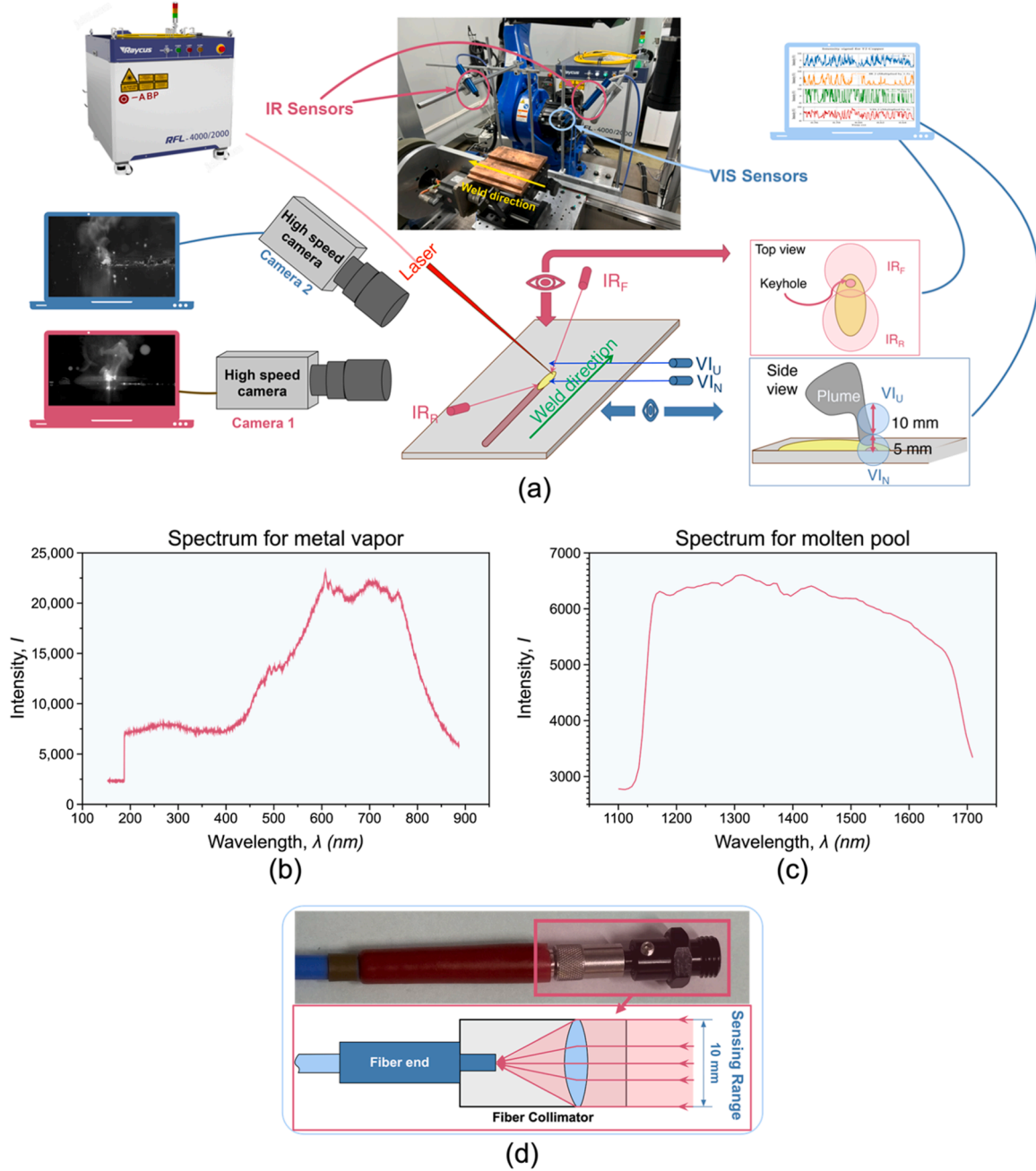


Fig. 1. (a) Schematic diagram of laser welding device and process monitoring system, containing two visible range photodiodes (VI_N and VI_U) and two infrared range photodiodes (IR_F and IR_R), capturing signals of metal vapor and molten pool, two high speed cameras taking image of molten pool and plume. The sensing range of all sensors were determined by collimators. (b) Experimental result of the spectrum for pure copper metal vapor. (c) Experimental result of the spectrum for molten pool. (d) Photo and schematic diagram of the structure of an optical sensing probe, using fiber collimator to limit sensing geometric range.

frame rate was 5000 frames per second. The resolution of image was 1280×800 .

3. Monitoring model

3.1. Objective and label

Due to the local light reflection of copper welds, the accuracy of visual inspection after welding is not ideal. To this end, a process monitoring system for copper laser welding is necessary. In the keyhole laser welding of pure copper, the metal evaporation in the lower part of keyhole is intensive. The oscillation of molten pool is obvious. The surface defects like pits or uneven toes were considered to be caused by accumulated energy inputs [18]. Therefore, a very short time of instability might not lead to a visible defect at seam. The prediction target of the models was divided into short-term unstable status and macroscopic surface defects.

For the model analyzing short-term stability states, the instabilities were determined using process high-speed images shown at left side in Fig. 2. To decide which stability states the image represents, which was needed in dataset building, a quantitative criterion was designed: In high-speed images, the length of molten pool and the height of oscillating liquid column could be measured using threshold image processing method. When the oscillation height larger than the half of molten pool length, the oscillation was considered to be in moderate

instability. When a visible melt ejection or a large spatter with diameter larger than half of the length of molten pool was found, the molten pool was considered in severe instability state. As shown in Fig. 2(a), slight instability was found to have minor impact on weld formation. As for moderate instabilities, they mostly resulted in small spatters and uneven width of weld, as shown in Fig. 2(b). This was unfavorable for the weld appearance but did not obviously damage the performance of the weld. For severe oscillation, large volume of liquid metal was thrown out from molten pool and deep pits were often found on the weld surface, as shown in Fig. 2(c). This attenuated the joint strength and the electrical conductivity, which was unacceptable in production application. Based on the criterion, the stability of copper welding process was scored referring to high-speed images, where 0.0 score of instability for no instability or slight instability, 0.5 score for moderate instability, and 1.0 score for severe instability. The short-term model was trained with these labeled scores.

For the target of predicting macroscopic surface defects, the output scores of the short-term model in a longer period were combined to represent macroscopic stability states of welding process. This model was trained with data labeled by weld surface formation topography and cross-section metallographic images shown as the center and right of Fig. 2. The labeled unacceptable defects included: (1) pits or undercuts with depth more than $1/3$ of weld penetration depth; (2) a change of seam width more than $1/3$ of mean width; (3) bulges with reinforcement more than 3 times of mean weld reinforcement. The target of the

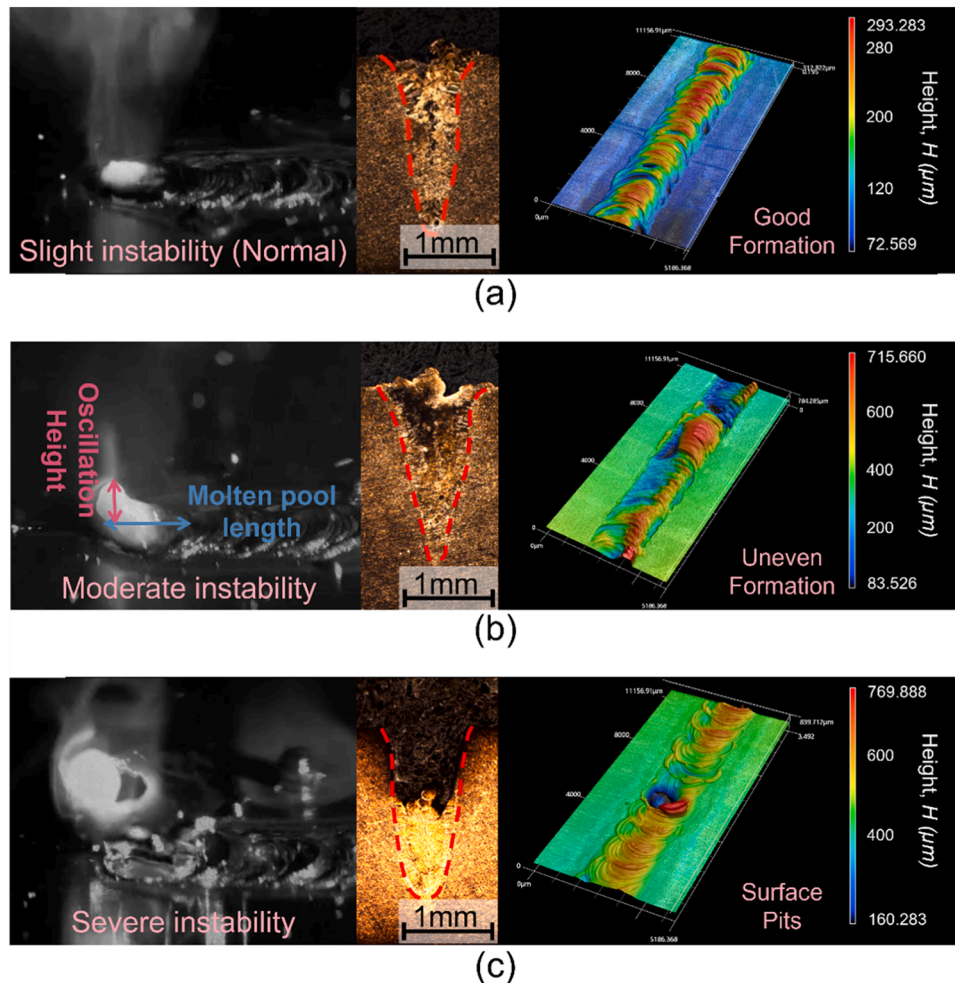


Fig. 2. The typical stability status of welding, with high-speed images of molten pool denoting stability status at left, cross-section seam metallographic photos at center, and surface topography at right. (a) Slightly oscillation and normal weld formation. (b) Moderate oscillation and uneven weld formation, where the definition criterion of oscillation height and molten pool length is marked. (c) Severe oscillation and surface pit.

macroscopic model was to predict these defects.

3.2. Spatial signals and feature extraction

For the laser welding of pure copper, keyhole and molten pool determine weld quality. The intensity and the change of optical radiation is closely related to the condition of keyhole and molten pool. Extracting features from optical signals and establishing a weld quality evaluation model is feasible. The temperature of vapor plume is very high. Its optical radiation includes both thermal radiation and atomic spectral lines. The wavelength mainly located in the visible band. When the keyhole was in the normal condition, the vapor plume erupted from keyhole, and gradually weakened in the upward process, as presented in Fig. 3(a). As the keyhole fluctuated periodically, the light intensity of the two photodiodes fluctuated synchronously. The light intensity detected by the VI_N was higher than that by the VI_U thanks to the vapor dissipation. When the melt ejections occurred, the molten pool fluctuated severely and resulted in the closure of keyhole opening. The light intensity of VI_N dropped significantly. However, the broken molten pool provided extra exit for the metallic vapor to escape [33]. Optical radiation detected by the upper photodiode became more intensive than that by the nether photodiode. This resulted in a reverse of the ratio between the VI_N and VI_U . Besides, the periodical change of the visible light signals was interrupted during the instability phenomena.

The temperature of copper molten pool is much lower than the vapor plume. The optical wavelength of its thermal radiation mainly located in the infrared band. When the molten pool was in the normal condition, the zone located near the laser incidence point had the highest temperature. The temperature at the rear of the molten pool decreased sharply and gradually solidified. The optical intensity detected by the IR_R was much lower than that by the IR_F , as shown in Fig. 3(b). When the instability of molten pool occurred, a large amount of liquid metal was pushed upwards the rear of the molten pool, or the liquid metal was directly lifted by the gas explosion [6]. This liquid flow increased the temperature or the volume of rear molten pool or, resulting in a rise of the IR_R . The optical intensity of the IR_R rapidly exceeded the IR_F . The

reversal of intensity ratio between the two infrared photodiodes normally last 2 ms to 4 ms.

Normally, the instable phenomena are transient, but the previous development processes have a certain time duration. The influence of feature extraction window duration on the model performance cannot be ignored [34]. To this end, the feature extraction was carried out in the window with a time duration of 1 millisecond. The four windows before the current window were considered as the development stage. The five windows formed a computation section, as presented in Fig. 4(a). The last window corresponded to the targeted moment. In each computation section, 14 features were extracted from the optical radiation collected by all photodiodes. Among them, the mean and standard deviation of infrared light intensity ($\overline{IR_F}$, $\overline{IR_R}$, σ_{IR_F} , and σ_{IR_R}), the mean value ratio between the visible light and the infrared light (r_{VI} and r_{IR}), and the fluctuation frequency peak position (f_{VI_N} and f_{VI_U}) and average intensity in the range from 100 Hz to 1000 Hz (keyhole frequency range, KFR, discussed in next section) of the visible light (I_{VI_N} and I_{VI_U}) were calculated inside the 1 ms window. The mean and standard deviation of infrared light intensity were calculated in the whole section ($\overline{IR_F}$, $\overline{IR_R}$, σ_{IR_F} , and σ_{IR_R}). All features were listed in Table 3.

To explore the frequency information of visible radiation, the Continuous Wavelet Transform (CWT) was carried out. The Morse wavelet was used as base wavelet to obtain the distribution of visible light signal in the time-frequency domain, as presented in Fig. 4(a). The fluctuation frequency of visible radiation mainly located within the range of 100 ~ 1000 Hz, which can be seen as the specific frequency range of keyhole oscillation (abbreviated as keyhole frequency range, KFR). Above this range, it indicates that the vapor pressure inside keyhole is too high. The intensive vapor eruption caused the rapid vibration of liquid metal at the open mouth of keyhole. Below this range, it means that the keyhole reconstruction time is too long. For the laser welding of pure copper, the multiple reflections of laser beam side the keyhole are the main ways to absorb laser energy. Normal oscillation of keyhole laid in the frequency range more than 500 Hz. The disappearance of keyhole for a long time will lead to the insufficient heat absorption and bad weld formation and caused abnormal fluctuation

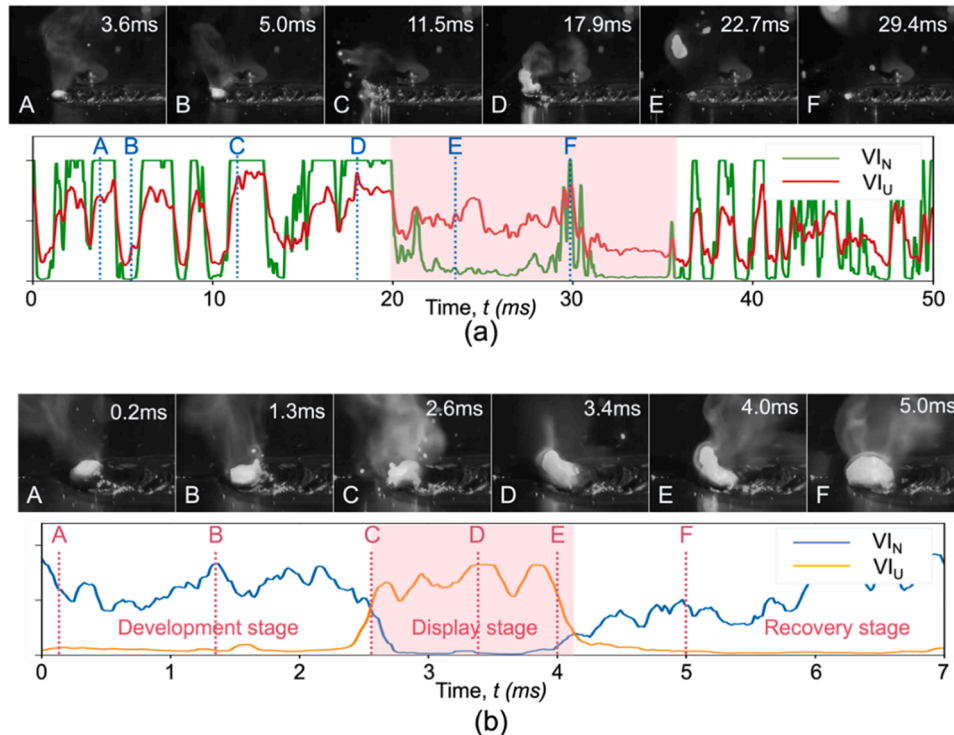


Fig. 3. Typical status of copper molten pool and corresponding optical signals. (a) Visible signals. (b) Infrared signals.

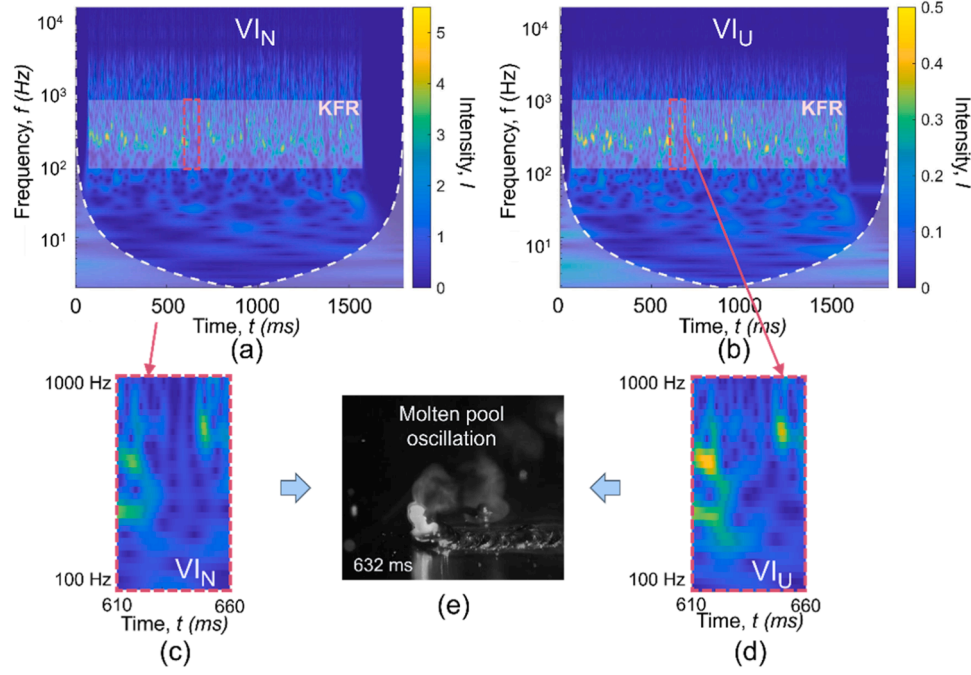


Fig. 4. The CWT results of VI signals and local signal at a typical molten pool oscillation point. (a) Global CWT result of VI_N . (b) Global CWT result of VI_U . (c) Local magnification of VI_N 's result at a molten pool oscillation point. (d) Local magnification of VI_U 's result at a molten pool oscillation point. (e) High-speed image of the molten pool oscillation point.

Table 3
the status features extracted in the laser welding process of pure copper.

Feature	Symbol	Description	Computation duration
P ₁	$\overline{IR_F}$	Mean intensity of IR_F signal	In targeted window (1 ms)
P ₂	σ_{IR_F}	Standard deviation of IR_F signal	
P ₃	$\overline{IR_R}$	Mean intensity of IR_R signal	
P ₄	σ_{IR_R}	Standard deviation of IR_R signal	
P ₅	r_{IR}	Intensity ratio between two IR signals	
P ₆	I_{VI_N}	Average intensity of VI_N in KFR	In whole Section (5 ms)
P ₇	I_{VI_U}	Average intensity of VI_U in KFR	
P ₈	r_{VI}	Intensity ratio between two VI signals in KFR	
P ₉	f_{VI_N}	Frequency peak position of VI_N signal	
P ₁₀	f_{VI_U}	Frequency peak position of VI_U signal	
P ₁₁	$\overline{IR_F}$	Mean intensity of IR_F signal	In whole Section (5 ms)
P ₁₂	σ_{IR_F}	Standard deviation of IR_F signal	
P ₁₃	$\overline{IR_R}$	Mean intensity of IR_R signal	
P ₁₄	σ_{IR_R}	Standard deviation of IR_R signal	

signal with lower frequency. Fig. 4(b) showed a typical keyhole instable process and its related CWT result. A high frequency peaks occurred during the process of melt ejection. The rapid increment of VI_U signal indicated a sharp eruption of metal vapor. In the process of melt ejection, normal keyhole fluctuation was interrupted, resulting in the loss of high frequency component for both results. However, low frequency peaks were found only in the VI_U result, which was caused by the large volume of spatters and metal vapor from the broken molten pool. Because the vapor plume at this point was not from the keyhole opening, such low frequency signals were not captured by VI_N sensor.

3.3. Multi-scale cascade model

The copper laser welding process exhibits significant fluctuation characteristics. Continuous unstable oscillation during welding process

would result in poor weld formation. Therefore, a dual model architecture was designed to predict the weld formation status. As presented in Fig. 5, there are a perception model and a decision model. The perception model takes signal features with short-term high-speed image labels and is in charge of monitoring the short-term stability of welding process. It outputs a stability score every 1 millisecond. The decision model predicts the weld defects based on the score array in the stack of the outputs of short-term perception model, which is a first-in first-out stack where the scores from the perception model are buffered in. An artificial neural network (ANN) with two hidden layers was used as the perception model. Its input is the features extracted from the optical radiation signals. The label comes from the stable state reflected by the high-speed photography. The labeling criterion was discussed in Section 3.1. The activation function is the ReLU function, and the loss function is mean square error. Learning rate is set to 0.0005. A support vector machine (SVM) with radial basis kernel function was used as the decision model. Its input is the stacked array of stability scores. The label comes from the surface formation of weld, where the unacceptable defects discussed in Section 3.1 with the classification target. The penalty coefficient C and the kernel function coefficient γ are set to 10 and 0.1. The length of the score array is set to 10. NumPy toolbox and MATLAB was used for signal processing and feature extraction. Scikit-learn toolbox and PyTorch library were used for model building.

For the perception model, there were total 20178 samples obtained from two groups of welding experiments. The number of samples with scores of 0, 0.5, and 1 are 9534, 8193 and 2451, respectively. The feature distribution was computed. The features related to the infrared radiation at the front of molten pool and the accumulated intensity of visible light radiated by metal vapor have a typical normal distribution. The features related to the infrared radiation at the rear of molten pool and the peak frequency of visible light radiated by metal vapor have a positive skewed distribution. It could be attributed to the molten pool movement to the rear end. These fluctuations make the signal distribution no longer symmetrical. Although the distribution skewness carries process information, it is disadvantageous for modelling. To this end, the Box-Cox transformation and the data scaling was executed [35]. After the data preparation, all features met the standard normal

Perception Model

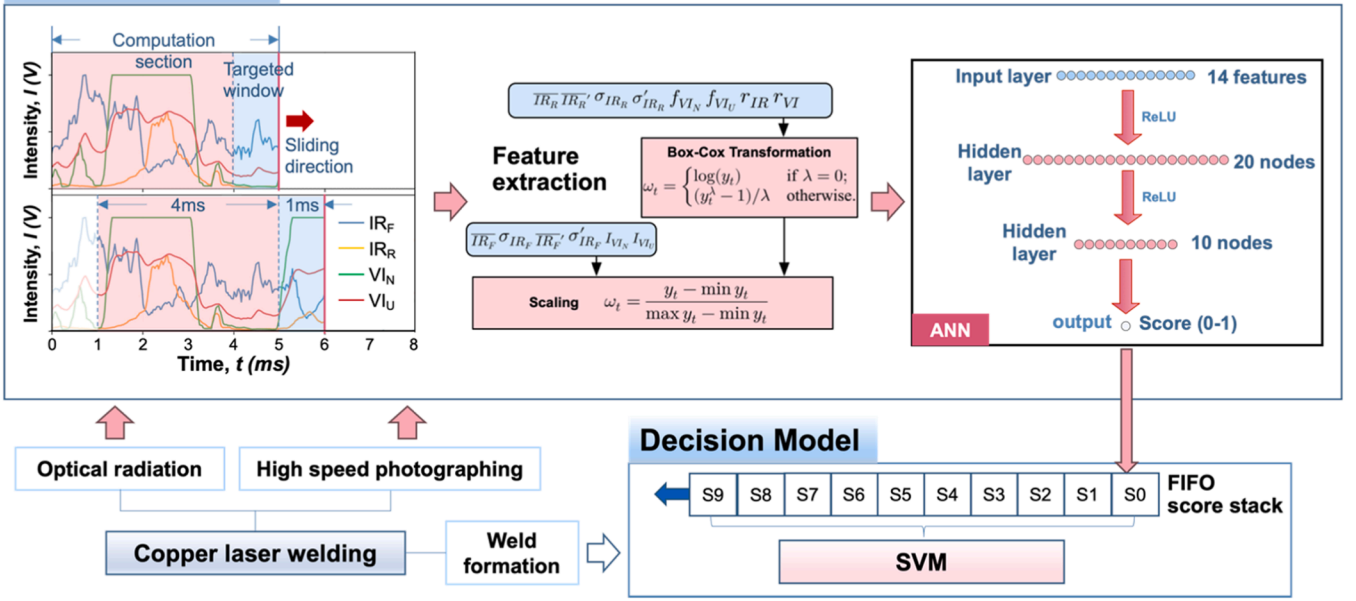


Fig. 5. The architecture of multi-scale cascade model for laser welding process stability monitoring. Features were calculated using sliding multi-scale windows. Original feature values were passed to a signal scaling preparation pipeline. The prepared features were used to train an ANN model to get short-term stability score. The stacked scores were passed to a SVM model to decide whether the process was stable.

distribution within similar scales.

To obtain the label of the decision model, the weld surfaces were scanned. The scanning interval is 0.05 mm. The height differences from the reference surface were computed on each scanning line. From this, the cumulative height deviation was obtained for a section of weld seam. For the group A, the section length was 0.5 mm since its welding speed was 3 m/min. For the group B, its welding speed increased to 6 m/min, the section length was 1 mm. In this study, the number of weld section with scores of 1 and 0.5 are 106 and 448. From the number of abnormal samples in two models, it can be found that the probability of process instability is high, but not all instability moments will lead to the poor

weld formation.

The two models were trained in sequence. The training set, validation set, and test set were selected randomly, and their proportions remained at 60 %, 20 %, and 20 %, respectively. The accuracy of perception model was evaluated based on score range thresholds for the three types of instabilities: (1) severe instability: >0.8 ; (2) moderate instability: $0.5-0.8$; (3) no instability or slight instability: <0.5 . The final accuracy was 92.3 %. The accuracy of decision model was 86.3 %. The total time consumption, including the feature extraction and the prediction of two models, was less than 500 ms.

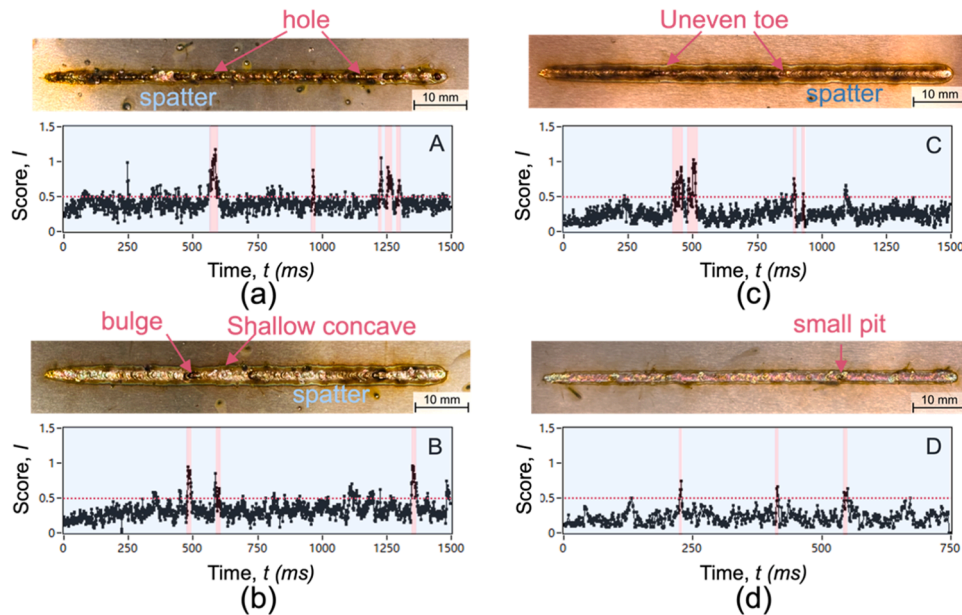


Fig. 6. The weld surface photos and the prediction results of stability based on the cascade model for the four test welds with a series of laser power settings. The predicted defective positions are marked in red. (a) Weld A (3 kW, 3 m/min). (b) Weld B (4.5 kW, 3 m/min), (c) Weld C (6 kW, 3 m/min). (d) Weld D (6 kW, 6 m/min).

4. Results and discussion

4.1. Model performance

A series of welding experiments were carried out according to the parameters listed in Table 2. The weld surface photos, as well as the results predicted by the cascade model, were presented in Fig. 6. According to the modeling mechanism, the closer the ANN score is to 0, the more stable the welding process is. If the scores exceed 0.5 and lasted for a period of time, it will be judged as abnormal by the SVM model. The welding speed of weld A, B, and C was the same at 3 m/min. Their laser power was 3 kW, 4.5 kW, and 6 kW, respectively. For the weld A shown in Fig. 6(a), the weld formation was poor. Its ANN scores were very close to the value 0.5. With the increment of laser power, the surface quality of weld was improved. The ANN scores decreased gradually. For the weld B presented in Fig. 6(b), the overall level of scores decreased. There was no significant change in the times and amplitude of scores exceeding 0.5. For the weld C shown in Fig. 6(c), the overall level of ANN scores further declined. But a section with high scores and long duration occurred. It indicated a severe oscillation of molten pool. It may be attributed to the high laser power. On the one side, the keyhole obtained more energy to maintain the inner vapor pressure. The molten pool oscillation caused by the shrinkage or collapse of keyhole could be suppressed. On the other hand, the vapor pressure may rise sharply once the input energy was accumulated inside the keyhole. The metallic vapor expansion or explosion will result in severe fluctuations in the molten pool. For the weld D shown in Fig. 6(d), it had the same heat input per unit length as the weld A. Both the laser power and the welding speed of the weld D were twice that of weld A. The ANN scores of weld D were much lower compared to the weld A. Although there were several times of score over 0.5, the magnitude and the duration were small. It could be inferred that the welding parameters of weld D was suitable for pure copper. The overall fluctuation of molten pool was kept in a small amplitude. At the same time, the severe explosion caused by the high accumulated energy was suppressed.

Statistical analysis was conducted on the ANN scores of welding processes with different parameters to evaluate the effectiveness of the model in monitoring global stability in a time period, and the results were presented in Fig. 7. The scores were divided into three sections using threshold of 0.5 and 0.8. As presented in Fig. 7(a), the proportion of scores exceeding 0.5 is close to 15 % in weld A. The proportion of weld B and weld C decreased obviously. But the proportion of scores exceeding 0.8 always existed. In weld D, the scores less than 0.5 dominated, and the proportion of scores over 0.8 disappeared. As shown in Fig. 7(b), the score distribution of all welds was close to the normal distribution pattern. From weld A to D, the peak shifted to the left gradually. The distribution of ANN scores was consistent with the

surface quality of welds. It indicates that the ANN model has a good perception ability for the welding process state.

The input data of ANN model were the features extracted from the spatial optical radiation. The distribution of features was presented in Fig. 8. Compared to the other three welds, the feature distribution of weld D was different to some extent, especially in P1, P5, P6 and P11. The differences deserved investigation since the surface morphology of weld D was the best. The features (P1, P5, P11) were extracted from infrared radiation, which reflected the molten pool directly. The first two features were extracted in 1 ms window. The P1 is the mean intensity of infrared radiation at the front end of molten pool. The P5 is the intensity ratio between two infrared sensors. For the weld D, the P1 mainly located at the range of high value while the P5 mainly located at the range of low value. It indicated that the molten pool mainly located at the vicinity of laser incident point. The probability of oscillation back to the rear end was relatively low. The feature P11 is the mean intensity of infrared radiation at the front end of molten pool but was extracted in 5 ms window. It had the similar distribution with P1. As the computation window widened, the concentration degree of data distribution increased. The overall difference between the weld D and other welds was kept. Relatively, the sensitivity of feature decreased with the increment of window width.

The feature (P6) was extracted from the visible-band radiation. It is the mean intensity of visible band radiation at the open of keyhole. When the open of keyhole was closed, the metallic vapor disappeared. For the weld D, the low-value proportion of P6 was very small. It could be inferred that the probability of keyhole closure in the weld D was much lower than other welds. The difference of P7 among welds was not as obvious as that of P6. P7 is the mean intensity of visible band radiation at 3 mm above the workpiece surface. After the metallic vapor escaped from the keyhole, it expanded and disappeared gradually. The inertia of P7 is higher than that of P6. If the time duration of keyhole closure is only several milliseconds, the change of P7 is not significant. Thus, it could be concluded that the spatial information was helpful to detect the state of molten pool and metallic vapor.

4.2. Oscillation of copper molten pool

In the copper laser welding, the oscillation of molten pool is a common phenomenon. It is demonstrated by the intensity change of two infrared sensors. The infrared radiation of the molten pool is determined by temperature and surface area. According to the results of the infrared thermometer, the mean temperature of the copper molten pool under the experimental conditions in this study was in the range of 1900–2200 °C. Normally, the temperature and area of the molten pool at the laser entrance are relatively high [32], resulting in a strong infrared radiation signal. At the rear of the molten pool, the infrared radiation gradually

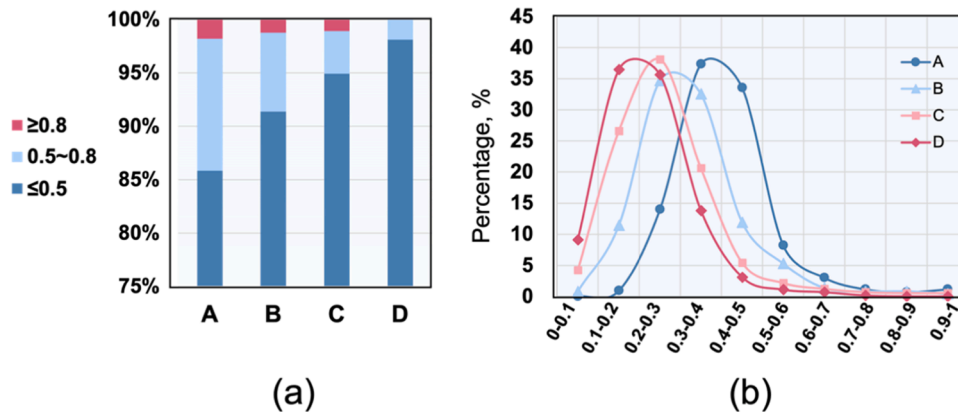


Fig. 7. The distribution and statistic result of ANN scores for four test welding groups. (a) Proportion of different score levels. (b) Distribution of score value levels in each group.

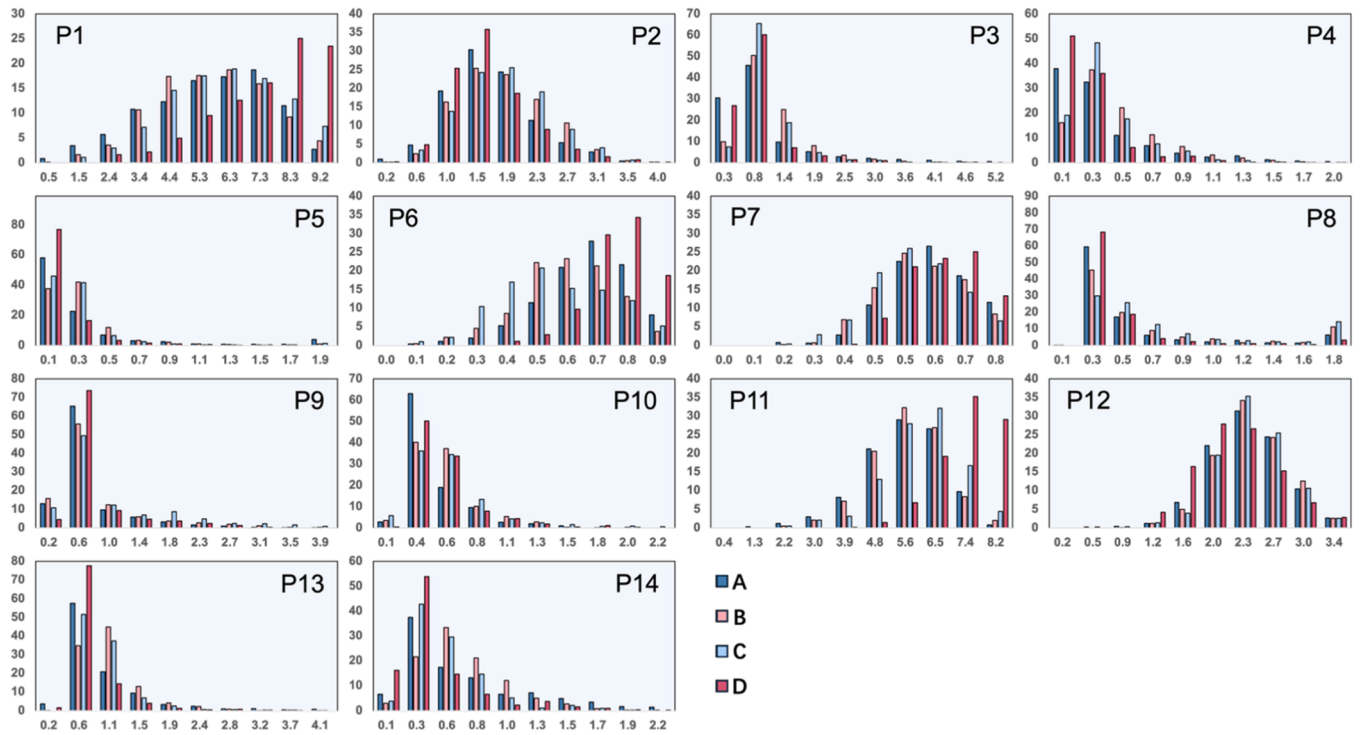


Fig. 8. The distribution of status feature values extracted in copper laser welding process for the four test welding groups.

weakened as the liquid metal cooled and solidified. During the molten pool oscillation, the high temperature liquid metal was pushed to the rear. The infrared radiation intensity detected by the rear photodiode increased, while the signal intensity of the front photodiode dropped. In Fig. 9(a), the photos of stable molten pool, as well as the optical signals, were presented. With the molten pool oscillated slightly backward, the IR_R intensity increased and the IR_F intensity decreased. The increment of IR_R intensity was small since the oscillation magnitude was small. The

oscillation of steady-state molten pool normally has the characteristics of small amplitude and high frequency. As shown in Fig. 9(b), the oscillation amplitude of the molten pool may be intensified. The amount of liquid metal participating in the oscillation increases. The time duration of one oscillation also increased. As long as the vast majority of liquid metal still falls back into the molten pool, this process will not cause significant changes in weld formation. The slight oscillation of molten pool is considered as stable process and their ANN scores are

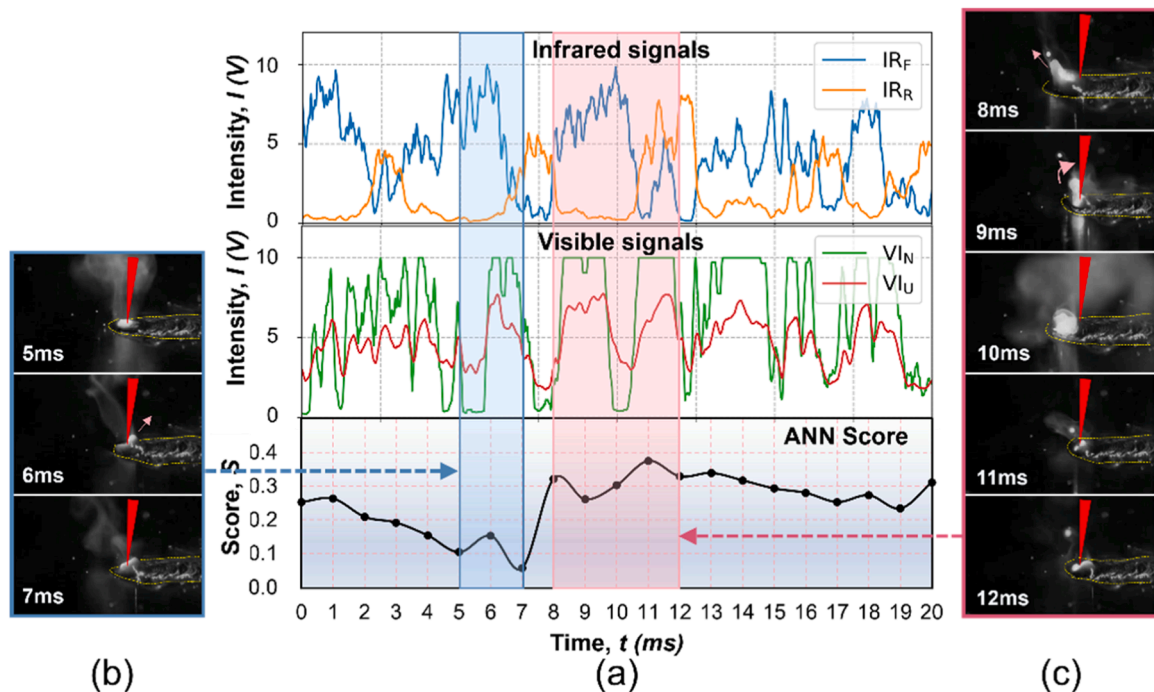


Fig. 9. The optical signals and ANN score of molten pool and related high-speed images in a typical slight oscillation situation. (a) Raw signals and ANN score. (b) High-speed image at stable status. (c) High-speed image at a slight oscillation point.

normally less than 0.4.

When the force on the molten pool changes, the oscillation amplitude increases. External forces may come from changes in welding conditions, or from expansion or collapse of keyholes. The more liquid metals participated in the oscillation, the greater the inertia of the molten pool was, and the frequency of oscillation decreased, as shown in Fig. 10 (a). The probability of liquid metal being thrown out and forming spatter increased. In Fig. 10 (b), a large liquid metal ball detached from the molten pool at the 11th millisecond. In the spatter formation process, the volume of molten pool dropped rapidly. The evaporation weakened accordingly. The intensity of both visible signals decreased. The intensity of infrared signal was affected by the trajectory of spatter. If the spatter moved near the front end of molten pool, its thermal radiation was detected by the IR_F sensor. The forward and backward movement of the molten pool, in which more liquid metal is involved, is called moderate oscillation. It may result in the large spatter or the uneven weld toes. The time duration is normally in the range from 10 ms to 20 ms. The ANN scores may exceed 1.0 but subside rapidly with the recovery of molten pool. The calculation results showed that the ANN model had good perception ability for this oscillation. Relatively, it is difficult to obtain the high accuracy of SVM model due to the short duration and the random direction of spatter.

4.3. Ejection of copper molten pool

In the laser welding of pure copper, the most severe instability of molten pool is the ejection of whole liquid metal. Due to the significant loss of metal, the ejection of molten pool will result in the deep pit on the weld surface. Both the resistivity and the strength of welded joints will be attenuated. Accordingly, this defect must be identified without false negatives. To this end, the optical signals of molten pool ejections were summarized for analysis. Normally, the severe instability has a development stage before outbreak, as presented in Fig. 11(a). In the development stage, the intrinsic frequency of molten pool oscillation changed. The liquid metal accumulated at the front end of molten pool. The intensity of visible radiation decreased which indicated the difficulty in escaping of metal vapor. After a few milliseconds of this process,

both the infrared signals and visible light signals sharply increased simultaneously. Both the IR_F signal and the VI_N signal exceeded the scale of sensors. As shown in Fig. 11(b), the whole molten pool was lifted by a blast. The volume of liquid metal thrown out is much larger than that of ordinary spatters. The keyhole disappeared with the molten pool. The interaction between laser beam and liquid metal was interrupted. The evaporation stopped. The intensity of VI_N signal dropped sharply and maintained at low level for several milliseconds. The severe instability is called the breakout stage of molten pool ejection. The evolution of this ejection process is revealed based on the photoelectric sensors distribute spatially around the molten pool. The ejection is transient, but the duration of the entire instable process is approximately 20 ms. During the process, the ANN scores kept over 0.8. It is favorable to obtain the high accuracy of SVM model. The experimental results showed that the detection ability of the cascade model for molten pool ejection reached 100 %.

4.4. Effectiveness analysis of signal representativeness

Molten pool ejection is a violent instability driven by the explosive force inside the molten pool. This phenomenon is particularly common in the laser welding of high-reflectivity materials like copper. It may be attributed to the low absorption of copper towards near-infrared laser. This instability is the main cause for surface defects like pits and uneven toe and is the primary prediction target of this work. By comparing the collected signals and molten pool dynamic, the representativeness of signal sensing and model prediction can be confirmed, and the understanding of the mechanism is enhanced.

Similar to a balloon explosion, the process of high-pressure bubbles inside the molten pool being punctured generates tremendous explosive force. The high-pressure bubbles were punctured due to energy concentration at keyhole bottom from the lack of absorption rate [18]. The huge bubble formed before the development stage. If taking a closer look at the beginning of melt ejection development, such process could be found both in high-speed images and captured signals, shown in Fig. 12. For the convenience of observation, the high-speed photography images were converted into pseudo-color images based on its temperature. At

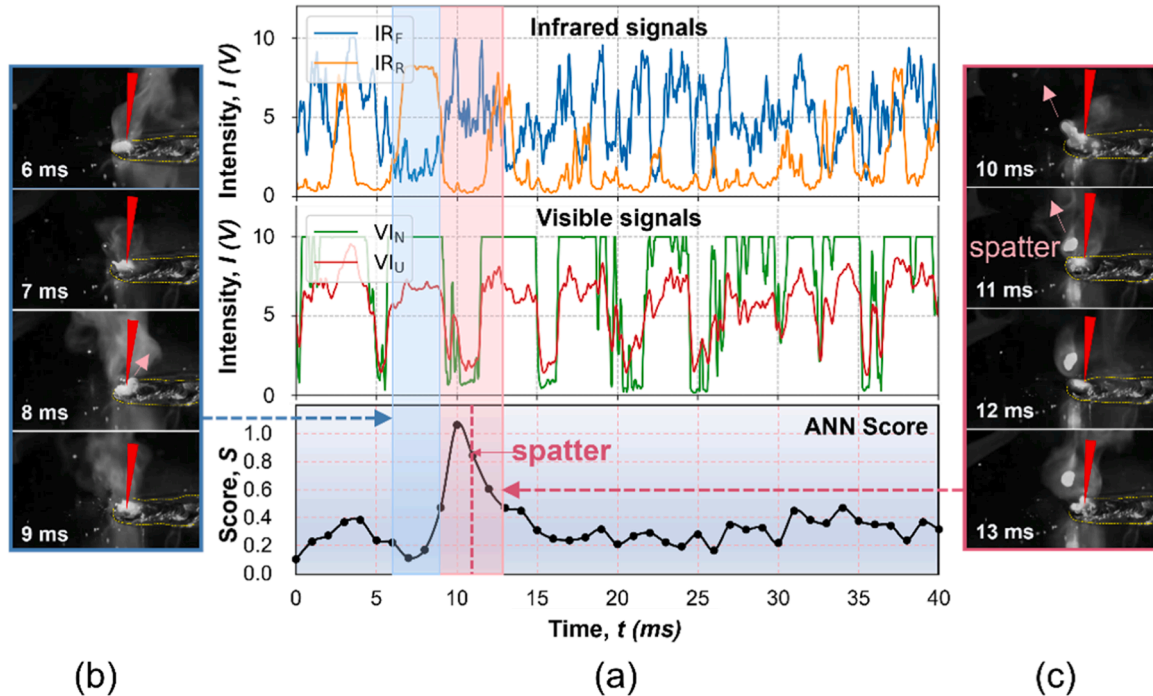


Fig. 10. The optical signals and ANN score of molten pool and related high-speed images in a spatter formation process. (a) Raw signals and ANN score. (b) High-speed image at spatter formation status. (c) High-speed images at spatter formation process.

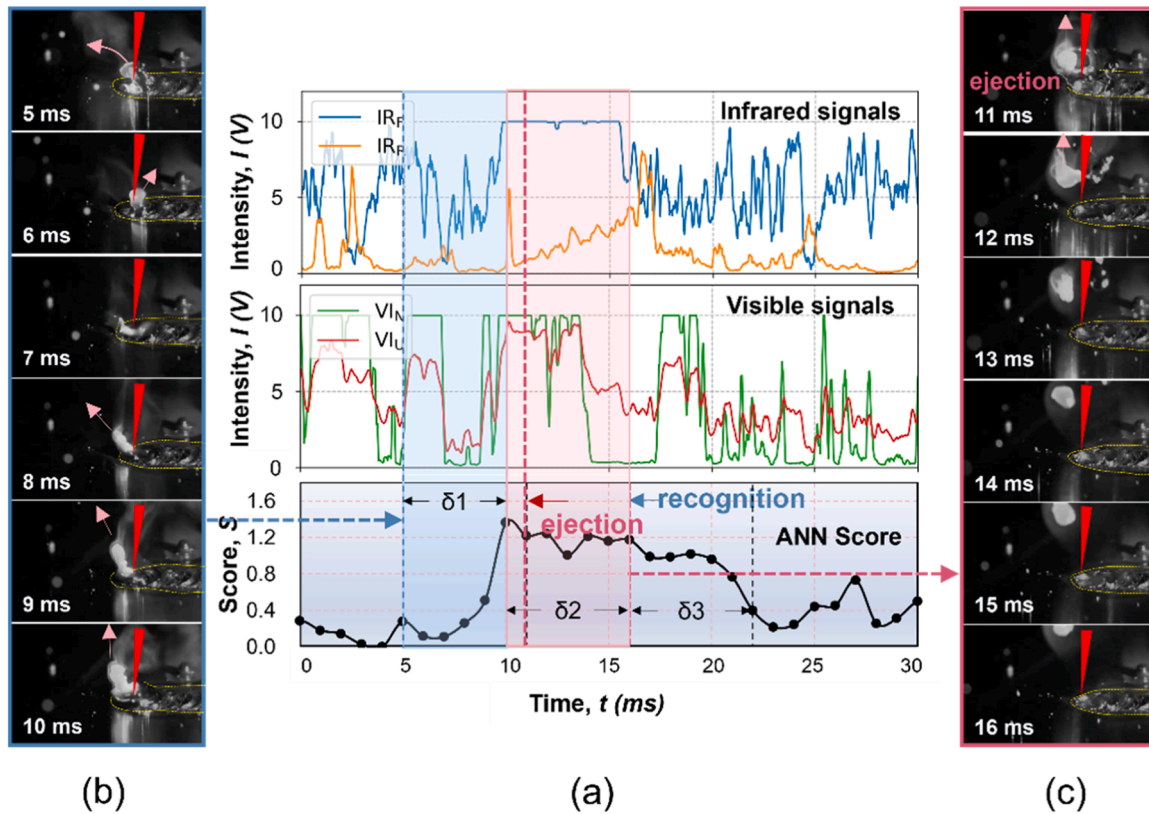


Fig. 11. The optical signals and ANN score of molten pool and related high-speed images at a melt ejection case. (a) Raw signals and ANN score. (b) Relative high-speed images at development stage of ejection. (c) Relative high-speed images in ejection process.

start, the liquid metal at the laser incidence position has the highest temperature when the molten pool was stable. The length of the molten pool was in the range from 1 mm to 3 mm. The temperature at the rear of molten pool decreased rapidly. The incident laser reflects multiple times inside a slender keyhole. Because of the low absorptivity, the laser energy absorbed by the upper part of the keyhole is relatively low. The laser transmitted energy to the sidewall through multiple reflections at the bottom of the keyhole. The lower part of the keyhole gradually expands as the evaporation increases. The change in keyhole morphology further increased the energy accumulation in the lower part. The evaporation at the bottom of the keyhole further intensified. With the intensive evaporation, the intensity of metal vapor outside the keyhole also increased. Under the force of keyhole expansion, the liquid metal was pushed towards the rear of the molten pool, with the rear of molten pool bulging obviously. The copper keyhole is prone to have both the narrow opening and the bloated bottom [33]. Such process caused abnormal oscillation of molten pool, which was captured by IR sensors. As is shown in Fig. 12, before the ejection occurred (time < 12 ms), the ratio of IR signals dramatically changed for multiple times. The half-period of oscillation was around 5 ms, which approximately equaled to the 5 ms window for IR feature extraction, while peak width of IR_R signal around 1 ms, which equaled to the 1 ms window. The multi-scale window extraction method effectively captured signal fluctuation at start stage in melt ejection. After that, the bottom of the keyhole continued to expand. The rear of molten pool continued to bulge. The stacked liquid metal may be collapsed under the action of gravity or laser beam, causing violent burst of molten pool, leading to a melt ejection at around 14 ms. At this time, for IR signals, intensity values of both signals decreased due to the loss of liquid metal. For VI signals, however, the burst at keyhole opening created high-frequency and high-amplitude fluctuation. By calculating frequency features from VI signals, this phenomenon could be detected by the model. After 17 ms, the disappearance and reconstruction phases of keyhole caused

the lack of VI signals and a continuous increase of IR_R signal values, which is similar to the case discussed in Section 4.3. From the analysis above, the monitoring target of all designed signal types were achieved, proving the reliability of the model, and the process can be further understood based on captured signal values.

5. Conclusion

A stability monitoring method for pure copper laser welding process has been constructed based on spatial photoelectric sensors and a cascade model. The main conclusions are as follows:

1. Optics signals in infrared range contains information of molten pool oscillation. The ratio of infrared signals at the front and back end of molten pool is closely related to oscillation of molten pool, which is important evidence to evaluate process stability. The multi-scale feature extraction window is designed to meet the half-period of molten pool oscillation and peak width of signals of abnormal oscillations. Statistical data with different feature extraction windows has different sensitivity for different instability phenomenon.
2. Optic signals in visible light range have a correlation with keyhole dynamic status and metal vapor emission, especially in frequency domain. With the help of Continuous Wavelet Transformation (CWT), it is found that the fluctuation of keyhole dynamics lies in a range of 100–1000 Hz, with normal frequency around 500 Hz. Different information exists at different height along metal vapor emission direction. When encountering keyhole collapse in melt ejection, high frequency peaks can be found at the start stage, the loss of high frequency component is found after that because of the interruption of keyhole normal fluctuation, and low frequency peaks are found in upper position above keyhole which is caused by the interference of spatters and large volume metal vapor from rear broken pool.

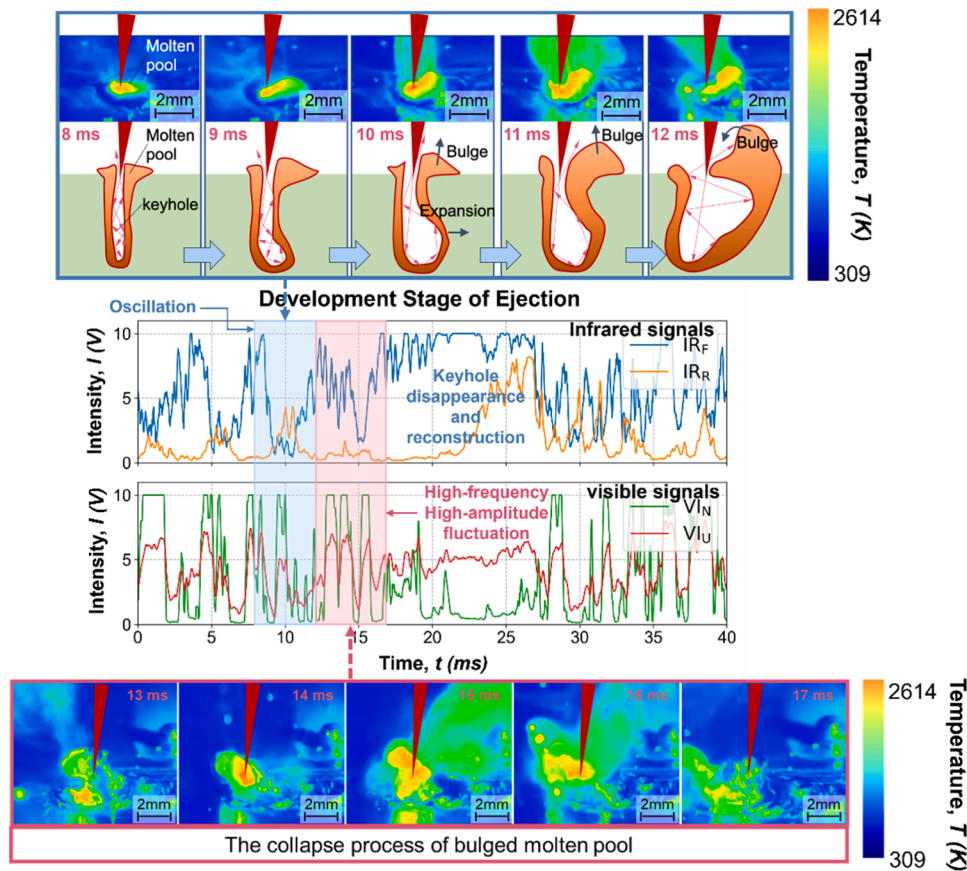


Fig. 12. Pseudo-color images of the development stage of a typical melt ejection process and their related signals.

3. A cascaded model was designed with a combination of a perception model and a decision model, based on Artificial Neural Network (ANN) and Support Vector Machine (SVM). The two-step model performs short-term status perception and macroscopic surface defects prediction in sequence. The separation of the monitoring of momentary instability and macroscopic defects is intended to enhance the model's adaptability of violent fluctuating signals. The test accuracy of the short-term perception model was 92.3 %, and the prediction accuracy of macroscopic decision model was 86.3 %, with severe unstable phenomenon like melt ejection detected without false negatives.
4. The mechanism of melt ejection developments met the phenomenon in signals. IR signals were able to detect liquid metal oscillation during the expansion of keyhole bottom, and VI signals captured abnormal high-frequency fluctuation of keyhole opening at the start of ejection. The outcome of sampled signals was consistent with system design. The representativeness of the signals was proved.

CRediT authorship contribution statement

Wucheng Li: Investigation. **Hao Dong:** Writing – original draft, Software, Resources, Investigation. **Yan Cai:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Weidong Mu:** Validation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Yan, S., Li, Z., Song, L., Zhang, Y., Wei, S., 2023. Research and development status of laser micro-welding of aluminum-copper dissimilar metals: a review. *Opt. Lasers Eng.* 161, 107312.
- [2] Wu, D., Zhang, P., Yu, Z., Gao, Y., Zhang, H., Chen, H., Chen, S., Tian, Y., 2022. Progress and perspectives of in-situ optical monitoring in laser beam welding: Sensing, characterization and modeling. *J. Manuf. Process.* 75, 767–791.
- [3] Kaerle, S., Overmeyer, L., Hoff, C., Herzog, P., Hermsdorf, J., 2016. Conditioning of copper material surfaces increasing the efficiency of continuous wave laser microwelding. *CIRP J. Manuf. Sci. Technol.* 14, 66–70.
- [4] Auwal, S.T., Ramesh, S., Yusof, F., Manladan, S.M., 2018. A review on laser beam welding of copper alloys. *Int. J. Adv. Manuf. Technol.* 96, 475–490.
- [5] Heider, A., Stritt, P., Hess, A., Weber, R., Graf, T., 2011. Process stabilization at welding copper by laser power modulation. *Phys. Procedia* 12, 81–87.
- [6] Baumann, R., Hipp, D., Lasagni, A.F., Mahrle, A., 2021. Tailored absorptivity: Increasing the laser weldability of copper through surface structuring. *Mater. Lett.* 283, 128700.
- [7] Franco, D., Oliveira, J.P., Santos, T.G., Miranda, R.M., 2021. Analysis of copper sheets welded by fiber laser with beam oscillation. *Opt. Laser Technol.* 133, 106563.
- [8] Kaufmann, F., Forster, C., Hummel, M., Olowinsky, A., Beckmann, F., Moosmann, J., Roth, S., Schmidt, M., 2023. Characterization of vapor capillary geometry in laser beam welding of copper with 515 nm and 1030 nm laser beam sources by means of in situ synchrotron X-ray imaging. *Metals* 13, 135.
- [9] Miyagi, M., Zhang, X., 2015. Investigation of laser welding phenomena of pure copper by x-ray observation system. *J. Laser Appl.* 27.
- [10] Zhang, L.-J., Zhang, G.-F., Ning, J., Zhang, X.-J., Zhang, J.-X., 2015. Microstructure and properties of the laser butt welded 1.5-mm thick T2 copper joint achieved at high welding speed. *Mater. Des.* 88, 720–736.
- [11] Dhara, S., Finuf, M., Zediker, M., Masters, I., Barai, A., Das, A., 2023. Utilising blue laser over infrared laser to enhance control of penetration depth and weld strength for producing electric vehicle battery interconnects. *J. Mater. Process. Technol.* 317, 117989.

- [12] Hess, A., Schuster, R., Heider, A., Weber, R., Graf, T., 2011. Continuous Wave Laser Welding of Copper with Combined Beams at Wavelengths of 1030nm and of 515nm. *Phys. Procedia* 12, 88–94.
- [13] Hummel, M., Kulkens, M., Schöler, C., Schulz, W., Gillner, A., 2021. In situ X-ray tomography investigations on laser welding of copper with 515 and 1030 nm laser beam sources. *J. Manuf. Process.* 67, 170–176.
- [14] Maina, M.R., Okamoto, Y., Hamada, K., Okada, A., Nakashiba, S., Nishi, N., 2022. Effects of superposition of 532 nm and 1064 nm wavelengths in copper micro-welding by pulsed Nd:YAG laser. *J. Mater. Process. Technol.* 299, 117388.
- [15] Kaufmann, F., Ermer, J., Maier, A., Roth, S., Schmidt, M., 2021. Influence of plume attenuation under high power laser welding of copper using visible wavelengths. *J. Laser Appl.* 33, 042006.
- [16] Ockel, M., Hensel, A., Franke, J., 2022. Suitability of a 2 kw 515 nm continuous wave laser for deep penetration welding of copper. 2022 12th Int. Electr. Drives Prod. Conf. EDPC 1–9.
- [17] Reisgen, U., Olschok, S., Jakobs, S., Turner, C., 2016. Sound Welding of Copper: Laser Beam Welding in Vacuum. *Phys. Procedia* 83, 447–454.
- [18] Ning, J., Zhang, L.-J., Yin, X., Zhang, J.-X., Na, S.-J., 2019. Mechanism study on the effects of power modulation on energy coupling efficiency in infrared laser welding of highly-reflective materials. *Mater. Des.* 178.
- [19] Zhang, Y., Chen, J., Zhang, W., Li, C., Qiu, C., Ding, J., Lu, H., Zhang, K., 2023. Study of spatter net forming mechanism and penetration mode under flexible ring mode laser welding. *J. Mater. Res. Technol.* 24, 2213–2225.
- [20] Kaufmann, F., Maier, A., Schrauder, J., Roth, S., Schmidt, M., 2024. Influence of laser beam intensity distribution on keyhole geometry and process stability using green laser radiation. In: Silva, F.J.G., Pereira, A.B., Campilho, R.D.S.G. (Eds.), *Flex. Autom. Intell. Manuf. Establ. Bridg. More Sustain. Manuf. Syst.* Springer Nature Switzerland, Cham, pp. 216–227.
- [21] Chianese, G., Hayat, Q., Jabar, S., Franciosa, P., Ceglarek, D., Patalano, S., 2023. A multi-physics CFD study to investigate the impact of laser beam shaping on metal mixing and molten pool dynamics during laser welding of copper to steel for battery terminal-to-casing connections. *J. Mater. Process. Technol.* 322, 118202.
- [22] Will, T., Kohl, J., Hoelbling, C., Müller, L., Schmidt, M., 2023. Laser welding of different pure copper materials under consideration of shielding gas influence and impact on quality relevant surface topographical features. *J. Laser Appl.* 35, 012013.
- [23] Sudo, M., Fujio, S., Takenaka, K., Mizutani, M., Sato, Y., Tsukamoto, M., 2023. Experimental study on pure copper welding with the 1.5kW blue diode laser for realization of a carbon-neutral society. In: Zediker, M.S., Zucker, E.P. (Eds.), *High-Power Diode Laser Technol.* XXI. SPIE, San Francisco, United States, p. 31.
- [24] Kamat, S., Cai, W., Rinker, T.J., Bracey, J., Xi, L., Tan, W., 2023. A novel integrated process-performance model for laser welding of multi-layer battery foils and tabs. *J. Mater. Process. Technol.* 320, 118121.
- [25] Xu, P., Ji, X., Li, M., Lu, W., 2023. Small data machine learning in materials science. *Npj Comput. Mater.* 9, 1–15.
- [26] Stavridis, J., Papacharalampopoulos, A., Stavropoulos, P., 2017. Quality assessment in laser welding: a critical review. *Int. J. Adv. Manuf. Technol.* 94, 1825–1847.
- [27] Lv, N., Zhong, J., Wang, J., Chen, S., 2014. Automatic measuring and processing system of audio sensing for real-time arc height control of pulsed GTAW. *Sens. Rev.* 34, 51–66.
- [28] Huang, Y., Xu, S., Yang, L., Zhao, S., Liu, Y., Shi, Y., 2019. Defect detection during laser welding using electrical signals and high-speed photography. *J. Mater. Process. Technol.* 271, 394–403.
- [29] Gao, X., Li, Z., Wang, L., Zhou, X., You, D., Gao, P.P., 2019. Detection of weld imperfection in high-power disk laser welding based on association analysis of multi-sensing features. *Opt. Laser Technol.* 115, 306–315.
- [30] Vakili-Farahani, F., Lungershausen, J., Wasmer, K., 2017. Wavelet analysis of light emission signals in laser beam welding. *J. Laser Appl.* 29, 022424.
- [31] Eren, B., Demir, M.H., Mistikoglu, S., 2023. Recent developments in computer vision and artificial intelligence aided intelligent robotic welding applications. *Int. J. Adv. Manuf. Technol.* 126, 4763–4809.
- [32] Chen, G., Zhang, M., Zhao, Z., Zhang, Y., Li, S., 2013. Measurements of laser-induced plasma temperature field in deep penetration laser welding. *Opt. Laser Technol.* 45, 551–557.
- [33] Heider, A., Sollinger, J., Abt, F., Boley, M., Weber, R., Graf, T., 2013. High-speed x-ray analysis of spatter formation in laser welding of copper. *Phys. Procedia* 41, 112–118.
- [34] Dong, H., Cai, Y., Li, Z., Hua, X., 2022. Multiscale feature extraction and its application in the weld seam quality prediction for plasma arc welding. *Int. J. Adv. Manuf. Technol.* 119, 2589–2600.
- [35] Box, G., Cox, D., 1964. An analysis of transformations. *J. R. Stat. Soc. Ser. B-Stat. Methodol.* 26, 211–252.